clusters and superclusters of galaxies. But the cosmos would double in size during its next half million years, and continue growing after that. At odds in the universe were two competing effects: gravity wants to make stuff coagulate, but the expansion wants to dilute it. If you do the math, you rapidly deduce that the gravity from ordinary matter could not win this battle by itself. It needed the help of dark matter, without which we would be living—actually not living—in a universe with no structures: no clusters, no galaxies, no stars, no planets, no people.

How much gravity from dark matter did it need? Six times as much as that provided by ordinary matter itself. Just the amount we measure in the universe. This analysis doesn't tell us what dark matter is, only that dark matter's effects are real and that, try as you might, you cannot credit ordinary matter for it.

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So dark matter is our frenemy. We have no clue what it is. It's kind of annoying. But we desperately need it in our calculations to arrive at an accurate description of the universe. Scientists are generally uncomfortable whenever we must base our calculations on concepts we don't understand, but we'll do it if we have to. And dark matter is not our first rodeo. In the nineteenth century, for example, scientists measured the energy output of our Sun and showed its effect on our seasons and climate, long before anyone knew that thermonuclear fusion is responsible for that energy. At the time, the best ideas included the retrospectively laughable suggestion that the Sun was a burning lump of coal. Also in the nineteenth century, we observed stars, obtained their spectra, and classified them long before the twentieth-century introduction of quantum physics, which gives us our understanding of how and why these spectra look the way they do.

Unrelenting skeptics might compare the dark matter of today to the hypothetical, now-defunct "aether" proposed in the nineteenth century as the weightless, transparent medium permeating the vacuum of space through which light moved. Until a famous 1887 experiment in Cleveland showed otherwise, performed by Albert Michelson and Edward Morley at Case Western Reserve University, scientists asserted that the aether must exist, even though not a shred of evidence supported this presumption. As a wave, light was thought to require a medium through which to propagate its energy, much as sound requires air or some other substance to transmit its waves. But light turns out to be quite